AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 08/April/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 1.5hrs

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

CamScanner

PART I: Answer the entire Multiple Choice Questions (MCQ). [15 Marks]

1) **Which of the following laws is fundamental to classical mechanics?**
A) Newton's Laws B) Einstein's Theory of Relativity
C) Planck's Law D) Faraday's Law

2) **In physics, a quantity is defined as:**
A) A mathematical expression B) Something that can be measured
C) A force or energy D) A dimensionless number

3) **What is the basic unit of length in the SI system?**
A) Meter B) Centimeter C) Kilometer D) Inch

4) **The position, x, of an object is given by the equation $x = A + Bt + Ct^2$, where t refers to time. What are the dimensions of A, B, and C?**
A) distance, distance, distance B) time, time, time
C) distance, time, time^2 D) distance, distance/time, distance/time^2

5) **Which of the following is the SI unit for electric current?**
A) Ampere B) Volt C) Ohm D) Watt

6) **The dimensions of the coefficient of friction are:**
A) [M L T^{-2}] B) [M L^2 T^{-2}] C) Dimensionless D) [L T^{-2}]

7) **Which of the following quantities is a vector?**
A) Mass B) Speed C) Velocity D) Energy

8) **In physics, a scalar quantity is:**
A) A quantity with direction B) A quantity that has magnitude and direction
C) A quantity that has magnitude only D) A quantity that changes with temperature

9) **If a vector $\vec{A}$ has components $A_X > 0$, and $A_Y < 0$, then the angle that this vector makes with the positive x-axis must be in the range**
A) 0° to 90° B) 90° to 180° C) 180° to 270° D) 270° to 360°

10) **The slope of a distance vs. time graph represents:**
A) Velocity B) Acceleration C) Displacement D) Force

11) **In dimensional analysis, if both sides of an equation have the same dimensions, the equation is:**
A) Dimensionless B) Incorrect C) Consistent D) Incomplete

12) **A car of mass 1000 kg is moving with an acceleration of 2 m/s^2. What is the net force applied on the car?**
A) 5000 N B) 2000 N C) 1000 N D) 200 N

13) **Which of the following is NOT a force?**
A) Tension B) Weight C) Acceleration D) Friction

14) **What force is responsible for keeping a satellite in orbit around the Earth?**
A) Tension B) Friction C) Gravitational force D) Electromagnetic force

15) **The radius of a circle is doubled. By what factor is the area changed?**
A) 8 B) 2 C) 4 D) 12

16) **The rate of change of displacement with respect to time is called:**
A) Velocity B) Speed C) Acceleration D) Distance

17) **When a force is called a "normal" force, it is**
A) the usual force expected given the arrangement of a system.
B) a force that is perpendicular to the surface of the Earth at any given location.
C) a force that is always vertical.
D) a contact force perpendicular to the contact surfaces between two objects.

18) **In a collision between two objects, the total momentum:**
A) Increases B) Decreases
C) Remains constant D) Depends on the masses of the objects

CamScanner

19) **A system of two connected objects is moving on a frictionless surface. If a force is applied, which of the following is true?**
A) The tension in the string remains constant
B) The acceleration of both objects is the same
C) The speed of both objects is the same
D) The force is equally divided between the two objects

20) **Impulse is defined as the:**
A) Change in momentum
B) Force applied over a period of time
C) Product of mass and acceleration
D) Rate of change of velocity

21) **A car traveling at 10.0 m/s has a constant acceleration of 2.0 m/s² in the same** direction as the velocity. After 2.0 seconds, the speed is
A) 10 m/s. B) 12 m/s. C) 8 m/s. D) 14 m/s.

22) **Which of the following is an example of a non-contact force?**
A) Tension in a rope B) Normal force C) Magnetic force D) Frictional force

23) **A moving object with constant acceleration. Its velocity-time graph will be:**
A) A horizontal line
B) A straight line passing through the origin
C) A straight line parallel to the x-axis
D) A curve

24) **The time for an object to hit the ground if dropped from a height of 78.4 m is:**
A) 4 seconds B) 3 seconds C) 5 seconds D) 6 seconds

25) **If the mass of an object is doubled while keeping its velocity constant, what happens to its momentum?**
A) It is halved B) It remains the same C) It is doubled D) It is quadrupled

26) **If a car moves with a momentum of 600 kg·m/s and the velocity is doubled, what will be the new momentum?**
A) 600 kg·m/s B) 1200 kg·m/s C) 300 kg·m/s D) 2400 kg·m/s

27) **The law of conservation of momentum is valid in which type of collision?**
A) Elastic collision
B) Inelastic collision
C) Perfectly inelastic collision
D) All of the above

28) **Which of the following is an example of a balanced force?**
a) Pushing a car uphill
b) Tug of war between two teams
c) A book resting on a table
d) Kicking a soccer ball

29) **The net force acting on an object is zero. Motion will be:**
A) It is at rest
B) It is moving with constant velocity
C) It is accelerating
D) It is impossible to determine without more information

30) **The force required to accelerate an object is directly proportional to its:**
A) Mass B) Velocity C) Acceleration D) Displacement

31) **A bullet shot straight up returns to its starting point in 10s. What is the initial speed of the bullet, assuming negligible air resistance?**
A) 9.8 m/s B) 25 m/s C) 49 m/s D) 98 m/s

32) **A ball collides with a second ball at rest. After collision, the first ball comes to rest and the second ball moves at the speed of the first ball. In this collision**
A) the masses are unequal and total kinetic energy is not conserved.
B) the masses are unequal and total kinetic energy is conserved.
C) the masses are equal and total kinetic energy is not conserved.
D) the masses are equal and total kinetic energy is conserved.

CamScanner

33) Four perpendicular Forces, A, B, C, and D. Force A direction is 45° north of east. Forces A and B both have a magnitude of 7.0 N, and vectors C and D both have a magnitude of 4.0 N. The sum of these four Forces is.

A) 4.2 N along the +y-axis
B) 3.5 N along + x axis
C) 4.2 N along the -y-axis
B) 3.5 N along - x axis

34) Which of the following quantities has units of a displacement?

A) 32 ft/s^2 downward
B) 40 km southwest
C) 9.8 m/s2
D) -120 m/s

35) Which of the following situations is impossible?

A) An object has velocity directed east and acceleration directed west.
B) An object has zero velocity but non-zero acceleration.
C) An object has constant non-zero acceleration and changing velocity.
D) An object has constant non-zero velocity and changing acceleration.

36) Under what condition is average velocity equal to the average of the object's initial and final velocity?

A) This can only occur if there is no acceleration.
B) The acceleration is constant.
C) This can occur only when the velocity is zero.
D) The acceleration must be constantly increasing

37) A rock from a volcanic eruption is launched straight up into the air with no appreciable air resistance. Which one of the following statements about this rock while it is in the air is correct?

A) On the way up, its acceleration is downward and its velocity is upward, and at the highest point both its velocity and acceleration are zero.
B) On the way down, both its velocity and acceleration are downward, and at the highest point both its velocity and acceleration are zero.
C) Throughout the motion, the acceleration is downward, and the velocity is always in the same direction as the acceleration.
D) The acceleration is downward at all points in the motion.

38) Ball A is dropped from the top of a building. One second later, ball B is dropped from the same building. Neglect air resistance. As time progresses, the difference in their speeds

A) increases.
B) remains constant.
C) decreases.
D) cannot be determined from the information given.

39) An object is moving with constant non-zero acceleration in the +x direction. The velocity versus time graph of this object is

A) a horizontal straight line.
B) a vertical straight line.
C) a straight line making an angle with the time axis.
D) a parabolic curve.

40) If the position versus time graph of an object is a horizontal line, the object is

A) moving with constant non-zero speed.
B) moving with constant acceleration.
C) at rest.
D) moving with increasing speed.

41) A motorist travels for 3.0 h at 80 km/h and 2.0 h at 100 km/h. What is her average speed for the trip?

A) 85 km/h
B) 88 km/h
C) 90 km/h
D) 92 km/h

42) A 50-kg box is resting on a horizontal floor. A force of 250 N directed at an angle of 30.0° below the horizontal is applied to the box. The coefficient of static friction between the box and the surface is 0.40, and the coefficient of kinetic friction is 0.30. What is the force of friction on the box?

A) 32 N
B) 200 N
C) 220 N
D) 250 N

CamScanner

43) A batter hits a foul ball. The 0.14-kg baseball that was approaching him at 40 m/s leaves the bat at 30 m/s in a direction perpendicular to the line between the batter and the pitcher. What is the magnitude of the impulse delivered to the baseball?

A) 9.8 N · s B) 1.4 N · s C) 3.5 N · s D) 7.0 N · s

44) In a closed system, Momentum is a conserved quantity?

A) True B) True only for elastic collisions
C) False D) Depends on object material

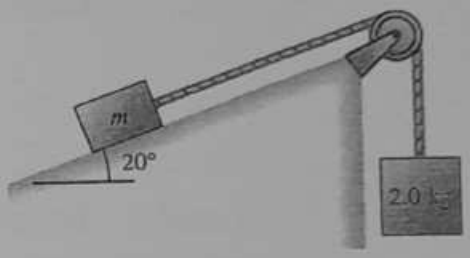
45) A 60-kg swimmer suddenly dives horizontally from a 150-kg raft with a speed of 1.5 m/s. The raft is initially at rest. What is the speed of the raft immediately after the diver jumps if the water has negligible effect on the raft?

A) 0.60 m/s B) 0.15 m/s C) 1.5 m/s D) 3 m/s

Part II: Answer the following questions. [15 Marks]

Question1: (5 marks)

The figure shows a block of mass m resting on a 20° slope. The block has coefficients of friction $\mu_s = 0.64$ and $\mu_k = 0.54$ with the surface of the slope. It is connected using a very light string over an ideal pulley to a hanging block of mass 2.0 kg. What is the minimum mass m so the system will remain at rest when it is released from rest?



Question2: (5 marks)

A toy rocket is launched vertically from ground level at time $t = 0.00$ s. The rocket engine provides constant upward acceleration during the burn phase. At the instant of engine burnout, the rocket has risen to 64 m and acquired an upward velocity of 60 m/s. The rocket continues to rise with insignificant air resistance in unpowered flight, reaches maximum height, and falls back to the ground. The time interval during which the rocket engine provided the upward acceleration, is closest to?

Question3: (5 marks)

An archer stands at rest on frictionless ice and fires a 0.50-kg arrow horizontally at 50.0 m/s. The combined mass of the archer and bow is 60.0 kg. With what velocity does the archer move across the ice after firing the arrow?



End of questions

CamScanner